PIETRO CORETTO

CURRICULUM VITAE ET STUDIORUM

Sunday 26th September, 2021

PERSONAL DATA

- **Address:** University of Salerno, Department of Economics and Statistics; Via Giovanni Paolo II, no.132, 84084, Fisciano (SA), Italy
- **Contacts:** e-mail: pcoretto@unisa.it; phone: (+39)-089-96.22.13; fax: (+39)-089-96.20.49

- **Citizenship:** Italian
- **Language:** Italian (mother tongue), English (fluent), French (basic)
- **Position:** Associate professor in statistics

EDUCATION

- **2004–2008:** Ph.D. in Statistical Sciences, University College London (UCL, UK). Primary supervisor: Christian Hennig (Department of Statistical Science, UCL). Secondary Supervisor: A. Philip Dawid (Department of Pure Mathematics and Mathematical Statistics, University of Cambridge)
- **2003–2004:** Master of Sciences in Economics, University of Pennsylvania (Philadelphia, Usa)
- **2002–2003:** Master in Economics and Finance (MEF), University of Napoli "Federico II" (Italy)
- **1998–2002:** Laurea in Economia e Commercio (B.A. in Economics and Business Studies), University of Salerno (Italy). Final score: summa cum laude with distinction. Thesis supervisor: Prof. Marco Pagano

HONORS, PRIZES AND SCHOLARSHIPS

- **2018:** gold medal for the "*Best Teaching Award*", MEF/LMEF graduate program in Economics and Finance, University of Naples "Federico II"
- **2009:** nominated for the *"Classification Society Distinguished Dissertation Award"* by the IFCS (International Federation of Classification Societies)
- **2007:** *"Costas Goutis Prize"*. For the best research work leading to the transfer to PhD registration. Awarded by the research committee of the Department of Statistical Sciences, University College London (UCL)
- **2006–2007:** *"Graduate School Scholarship"*, from University College London (UCL)
- **2003–2005:** *"Giovanna Crivelli"* scholarship from Unicredito Italiano Spa
- **2002:** *"Angelo Costa Prize"* (VI edition, International). Best of the papers selected by an international scientific committee comprising R. Solow (Nobel 1987), R. Mundell (Nobel 1999), P. Diamond, et al.
- **2002:** Banca Nazionale delle Comunicazioni Prize (for high quality BA dissertations selected in Italy). From B.N.C. Foundation
- **2002–2003:** Scholarship from Italian Society for Investment Professionals (ISIP)

PROFESSIONAL EXPERIENCES AND RESEARCH PROJECTS

- **2006–present:** Principal investigator in FARB (local) research projects
- **2013–2017:** Research collaborator, project *"A multicriterion approach for cluster validation"* (EP/K033972/1) and funded by Engineering and Physical Sciences Research Council (EPSRC, UK). Funded 98,024 GBP.
- **2012–2015:** Research collaborator, project (PRIN 2010J3LZEN) *"Forecasting economic and financial time series: understanding the complexity and modeling structural change"*, founded by Ministero dell'Istruzione, dell'Università e della Ricerca (MIUR). Funded: 506,134 Euro.
- **2009–2012:** Research collaborator, project *"High Performance Computing"*, funded by University of Salerno (ASSA098434)
- **2008–2011:** Research collaborator, project *"Evolutionary Computation in Statistics"* (PRIN 20077P5AWA), founded by Ministero dell'Istruzione, dell'Università e della Ricerca (MIUR). Funded 52,315 Euro.
- **2007–2009:** Research collaborator, project *"Economic consequences of the demographic transition in Europe"* (INTERLINK II04C0ECL72007). Partners: Institute of Fiscal Studies (IFS, London), Tilburg University, Goethe University Frankfurt.
- **2006:** Consultancy for Cirrus Capital, London, UK
- **2005–2006:** Participation to the project: *"Quantitative Analysis of Financial Markets"*, University of Salerno
- **2002–2002:** Research assistant for the international project *"Financial Market Integration, Corporate Financing and Economics Growth"*. Funded by the European Commission

SERVICE TO PROFESSION

Organization of Scientific Events

- **2018:** member of organizing committee of the "*International Summer School on Clustering, Data Analysis and Visualization of Complex Data*"; organized by ERS-IASC, ECAS and SIS-CLADAG; May 21–25; Catania, Italy
- **2018:** member of the local organizing committee for the "*4th Conference of the International Society for Nonparametric Statistics*" (ISNPS2018), June 11–15 2018, Salerno, Italy
- **2016:** member of the local organizing committee for the "*48th Meeting of the Italian Statistical Society*" (SIS2016), June 8–10 2016, Salerno, Italy
- **2014:** member of the local organizing committee for the "*6th International Conference on Mathematical and Statistical Methods for Actuarial Sciences and Finance*" (MAF2014), April 22–24, 2014, Vietri sul Mare (SA) - Italy
- **2010:** member of the local organizing committee for the "*4th International Conference on Mathematical and Statistical Methods for Actuarial Sciences and Finance*" (MAF2010), April 7–9, 2010, Ravello (SA) - Italy

Editorial Activity. Referee for (in alphabetical order): Advances in Classification and Data Analysis, Bioinformatics, Biometrika, BMC Bioinformatics, Computational Statistics and Data Analysis, Communications in Statistics, Electronic Journal of Statistics, IEEE Transactions on Cybernetics, IEEE Transactions on Pattern Analysis and Machine Intelligence, Journal of Computational and Graphical Statistics, Journal of Machine Learning Research, Journal of Multivariate Analysis, Journal of the Korean Statistical Society, Mathematical and Computational Applications, Metron, Statistical Methods and Applications, Statistical Modelling, Statistical Papers, Statistics and Computing.

Local administrative duties

- **2012–present:** member of the academic board of the PhD program in *Economics and Policy of Markets and Firms*, University of Salerno

- **2018–present:** member of committee for the project "Piano Nazionale Lauree Scientifice" (PLS, founded by MIUR), University of Salerno
- **2018–present:** member of the committee for student orientation; coordinator for the degree in *Statistics for Big Data*, University of Salerno
- **2012–present:** member of the editorial committee; Department of Economics and Statistics, University of Salerno

INVITED TALKS CONFERENCES, WORKSHOPS AND SEMINARS

- Seminar at Department of Statistics "P. Fortunati", University of Bologna, April 1, 2021, Bologna, Italy
- International Conference on Mathematical and Statistical Methods for Actuarial Sciences and Finance (MAF2020), April 15–17, 2010, Geneva, Switzerland
- CMStatistics 2019, "*13th International Conference of the ERCIM WG on Computational and Methodological Statistics*", December 19–21, 2020, London, UK
- CLADAG 2019, "*12-th Scientific Meeting Classification and Data Analysis Group*" Cassino, September 11–13, 2019, Cassino, Italy
- MBC^2-2018, conference on "*Model-Based Clustering and Classification*", September 5–7, 2018, Catania, Italy
- IFCS-2017, Conference of International Federation of Classification Societies, August 8–10, 2017, Tokyo, Japan
- SIS2017, "*Statistics and Data Science: New challenges, new generations*", Conference of the Italian Statistical Society (SIS), June 28-30, 2017, Firenze, Italy
- Workshop on Model-Based Clustering and Classification (MBC2), Università di Catania, Italy, September 2016
- 10th Scientific Meeting of the CLADAG Group of the Italian Statistical Society, Santa Margherita di Pula, Italy, October 2015
- 2nd Workshop PRIN 2010J3LZEN group, Università Bologna, Italy, November 2015
- 7th International Conference of the ERCIM WG on Computational and Methodological Statistics, University of Pisa, Italy, December 2014
- Workshop on Model-Based Clustering and Classification (MBC2), University of Catania, Italy, September 2014
- 1st Workshop PRIN 2010J3LZEN, University of Tor Vergata, Rome, Italy, June 2014
- 47th Scientific meeting Italian Statistical Society (SIS), Cagliari, Italy, June 2014
- 6th International Conference of the ERCIM WG on Computing and Statistics, University of London, London, UK, December 2013
- 46th Scientific meeting Italian Statistical Society (SIS) Rome, Italy, June 2012
- 4th International Conference of the ERCIM WG on Computing and Statistics, University of London, London, UK, December 2011
- International conference on Evolutionary Computation in Statistics, University of Tor Vergata, Roma, Italy, June 2011
- 3rd International Conference of the ERCIM WG on Computing and Statistics, University of London, London, UK, December 2010
- International Conference on Mathematical and Statistical Methods for Actuarial Sciences and Finance (MAF10), Ravello, Italy, April 2010
- International conference on Evolutionary Computation in Statistics, University Ca' Foscari, Venezia, Italy, March 2010
- 7th Scientific Meeting of the CLADAG Group of the Italian Statistical Society, Catania, Italy, September 2009

- JSM-ASA, Joint Statistical Meeting of the American Statistical Association, Washington DC, USA, August 2009
- Model-Based Clustering Summer Session Workshop, Institut Henri Poincaré, Paris, France, July 2009
- Seminar at Dpto. Estadística e Investigación Operativa, Universidad de Valladolid, Valladolid, Spain, November 2008.
- 1st Workshop of the ERCIM Working Group on Computing & Statistics, Neuchâtel, Switzerland, June 2008
- 1st joint meeting of the Société Francophone de Classification and the Classification and Data Analysis Group (CLADAG) of the Italian Statistical Society, Caserta, Italy, June 2008
- International Conference on Mathematical and Statistical Methods for Actuarial Sciences and Finance (MAF08), Venice, Italy, March 2008
- 6th Scientific Meeting of the Classification and Data Analysis Group of the Italian Statistical Society, Macerata, Italy, September 2007
- 31st Annual Conference of the German Classification Society on Data Analysis, Machine Learning, and Applications, Freiburg, Germany, March 2007
- seminar, Department of Statistical Sciences, University College London, London, UK, September 2006
- 29th Research Students' Conference in Probability and Statistics (RSC 2006), Glasgow, UK, April 2006
- 28th Research Students' Conference in Probability and Statistics (RSC 2005), Cambridge, UK, April 2005

TEACHING

Current courses

- Econometrics (graduate); *Master in Economics and Finance* (MEF), level II master degree, University of Naples Federico II (Since 2016)
- Advanced Statistical Learning (graduate); degree in *Statistics for Finance/Big Data Analytics*; University of Salerno (Since 2018)
- Data Science (graduate); PhD in *Economics and Policy of Markets and Firms*, University of Salerno (Since 2015)
- Statistical Learning (undergraduate); degree in *Statistics for Big Data*; University of Salerno (Since 2018)
- Data Analysis and Visualization (undergraduate); degree in *Statistics for Big Data*; University of Salerno (Since 2018)
- Computational Statistics (undergraduate); degree in *Management and Computer Science*; University of Salerno (Since 2009)

Past courses

- **2018:** lectures on Advanced Topics in Cluster Analaysis (graduate level) at the "*International Summer School on Clustering, Data Analysis and Visualization of Complex Data*"; organized by ERS-IASC, ECAS and SIS-CLADAG; Catania (Itlay), May 21-25
- **2009–2019** Statistics (undergraduate); degree in *Management and Computer Science*; University of Salerno
- **2013–2019** Probability and Inference (graduate); *Master in Economics, Finance and Risk Management* (MEFiRM), level I master degree, University of Salerno
- **2013–2014:** Mathematical Statistics (graduate level); PhD in "Economics, Law and Statistics for Policy, Markets, and Firms", University of Salerno
- **2008–2013:** Mathematical Statistics (graduate level); PhD in "Engineering and Economics of Innovations", University of Salerno

- **2008–2009:** Sampling Theory and Methods for Business Applications (undergraduate level); Faculty of Economics, University o Salerno
- **2006–2008:** Computational Statistics (graduate level); PhD in "Engineering and Economics of Innovations", University of Salerno
- **2006–2008:** Data Analysis (undergraduate level); Faculty of Economics, University of Salerno
- **2008:** Statistics (undergraduate level); GO.TER, II level Master (equivalent to MA), University of Salerno

RESEARCH PUBLICATIONS

Papers in journals (with peer-review)

Coretto, P. (2002). A theory of decidibility: Entropy and choice under uncertainty. *Rivista di Politica Economica*, vol. 92(11/12): pages 29–62.

Coretto, P. and C. Hennig (2010). A simulation study to compare robust clustering methods based on mixtures. *Advances in Data Analysis and Classification*, vol. 4(2): pages 111–135. URL: https://www.doi.org/10.1007/s11634-010-0065-4.

Coretto, P. and C. Hennig (2011). Maximum likelihood estimation of heterogeneous mixtures of gaussian and uniform distributions. *Journal of Statistical Planning and Inference*, vol. 141(1): pages 462–473. URL: https://www.doi.org/10.1016/j.jspi.2010.06.024.

Coretto, P. (2013). Discussion (invited) of: How to find an appropriate clustering for mixed-type variables with application to socio-economic stratification; by C. Hennig and T. F. Liao. *Journal of the Royal Statistical Society, Series C (Applied Statistics)*, vol. 62: pages 346–347. URL: https://doi.org/10.1111/j.1467-9876.2012.01066.x.

Coretto, P. and C. Hennig (2016). Robust improper maximum likelihood: tuning, computation, and a comparison with other methods for robust gaussian clustering. *Journal of the American Statistical Association*, vol. 111(516): pages 1648–1659. URL: https://www.doi.org/10.1080/01621459.2015.1100996.

Coretto, P. and F. Giordano (2017). Nonparametric estimation of the dynamic range of music signals. *Australian & New Zealand Journal of Statistics*, vol. 59(4): pages 389–412. URL: https://www.doi.org/10.1111/anzs.12217.

Coretto, P. and C. Hennig (2017). Consistency, breakdown robustness, and algorithms for robust improper maximum likelihood clustering. *Journal of Machine Learning Research*, vol. 18(142): pages 1–39. URL: http://jmlr.org/papers/v18/16-382.html.

Serra, A., P. Coretto, M. Fratello, and R. Tagliaferri (2018). Robust and sparse correlation matrix estimation for the analysis of high-dimensional genomics data. *Bioinformatics*, vol. 34(4): pages 625–634. URL: https://www.doi.org/10.1093/bioinformatics/btx642.

Coretto, P., A. Serra, and R. Tagliaferri (2018). Robust clustering of noisy high-dimensional gene expression data for patients subtyping. *Bioinformatics*, vol. 34(23): pages 4064–4072. URL: https://www.doi.org/10.1093/bioinformatics/bty502.

Serra, A., S. Önlü, P. Coretto, and D. Greco (2019). An integrated quantitative structure and mechanism of action-activity relationship model of human serum albumin binding. *Journal of Cheminformatics*, vol. 11(38): pages 1–10. URL: https://www.doi.org/10.1186/s13321-019-0359-2.

Coretto, P. (2020). Discussion (invited) of: Multiple-systems analysis for the quantification of modern slavery: classical and bayesian approaches; by Bernard W. Silverman. *Journal of the Royal Statistical Society, Series A (Statistics in Society)*, vol. 183(3): pages 725–726. URL: https://www.doi.org/10.1111/rssa.12505.

Storti, G. and P. Coretto (2020). Discussion (invited) of: Linear mixed effects models for non-gaussian continuous repeated measurement data; by O. Asar, D. Bolin, P. J. Diggle and J. wallin. *Journal of the Royal Statistical Society, Series C (Applied Statistics)*, vol. 69(5): page 46. URL: https://doi.org/10.1111/rssc.12405.

Coretto, P., M. La Rocca, and G. Storti (2020). Improving many volatility forecasts using cross-sectional volatility clusters. *Journal of Risk and Financial Management*, vol. 13(4): pages 1–23. URL: https://www.doi.org/10.3390/jrfm13040064.

Giordano, F. and P. Coretto (2020). A monte carlo subsampling method for estimating the distribution of signal-to-noise ratio statistics in nonparametric time series regression models. *Statistical Methods & Applications*, vol. 29(3): pages 483–514. URL: https://www.doi.org/10.1007/s10260-019-00487-5.

Hennig, C. and P. Coretto (2021). An adequacy approach for deciding the number of clusters for otrimle robust gaussian mixture-based clustering. *Australian & New Zealand Journal of Statistics*, pages 1–25. URL: https://doi.org/10.1111/anzs.12338.

Coretto, P. (2021). Estimation and computations for gaussian mixtures with uniform noise under separation constraints. *Statistical Methods & Applications*, pages 1–32. URL: https://doi.org/10.1007/s10260-021-00578-2.

Papers in book collections (with peer-review)

Hennig, C. and P. Coretto (2008). The noise component in model-based cluster analysis. In C. Preisach, H. Burkhardt, L. Schmidt-Thieme, and R. Decker, editors, *Data Analysis, Machine Learning and Applications*, pages 127–138. Springer Berlin Heidelberg, Berlin, Heidelberg.

Coretto, P. and M. L. Parrella (2010). Empirical likelihood based nonparametric testing for capm. In M. Corazza and C. Pizzi, editors, *Mathematical and Statistical Methods for Actuarial Sciences and Finance*, pages 103–112. Springer Milan, Milano.

Coretto, P., M. La Rocca, and G. Storti (2011). Group structured volatility. In S. Ingrassia, R. Rocci, and M. Vichi, editors, *New Perspectives in Statistical Modeling and Data Analysis*, pages 329–335. Springer, Heidelberg Dordrecht London New York.

Coppola, G. and P. Coretto (2012). Classificazione del territorio provinciale. In A. VV., editor, *Analisi delle caratteristiche strutturali e funzionali delle imprese industriali operanti in provincia di Salerno. Settore Alimentare e Chimico Plastico*, pages 43–68. Edizioni Scientifiche Italiane, Napoli.

Amendola, A., P. Coretto, and G. Storti (2013). Indagine campionaria. In AA.VV., editor, *Analisi delle Caratteristiche Strutturali e Funzionali delle Imprese Industriali Operanti in Provincia di Salerno: Settore Costruzioni*, pages 39–65. Edizioni Scientifiche Italiane, Napoli.

Coretto, P., M. La Rocca, and G. Storti (2021). A garch–type model with cross-sectional volatility clusters. In M. Corazza, M. Gilli, C. Perna, C. Pizzi, and M. Sibillo, editors, *Mathematical and Statistical Methods for Actuarial Sciences and Finance*, pages 1–6. Springer International Publishing.

Papers in conference proceedings (with peer-review)

Coretto, P. and C. Hennig (2007). Choice of the improper density in robust improper ML for finite normal mixtures. In *Classification and Data Analysis 2007*, pages 439–442. Springer, Hildeberg.

Coretto, P. (2008). Robust mixture modelling. In *Joint Meeting of the SFC and CLADAG*, pages 69–72. Edizioni Scientifiche Italiane, Napoli.

Coretto, P., M. La Rocca, and G. Storti (2009). Group structured volatility. In *Book of Short Papers CLADAG09 Catania*, pages 119–122. CLEUP, Padova.

Coretto, P. and C. Hennig (2009). Robust model-based clustering using an improper constant density selected by a distance optimizatin for non-outliers. In *JSM Proceedings, IMS General Theory Section*, pages 3847–3858. American Statistical Association, Alexandria, VA.

Coretto, P. (2012). Adaptive robust location–scale estimation. In S. I. di Statistica, editor, *Proceedings of the XLVI Scientific Meeting*, pages 1–4. Cleup, Roma.

Coretto, P. and C. Hennig (2014). Robust covariance matrix estimation with regularization. In S. I. di Statistica, editor, *Proceedings of the 47th Scientific Meeting of the Italian Statistical Society*, pages 1–8. Cuec, Cagliari.

Coretto, P. and C. Hennig (2015). Robust model-based clustering with covariance matrix constraints. In *10th meeting of Classification and Data Analysis Group*, pages 1–5. CUEC Editrice, Cagliari.

Serra, A., P. Coretto, and R. Tagliaferri (2017). On the noisy high-dimensional gene expression data analysis. In A. Petrucci and R. Verde, editors, *Statistics and data science: new challenges, new generations*, pages 919–926. Firenze University Press, Firenze.

Coretto, P., A. Serra, and R. Tagliaferri (2019). Noise resistant clustering of high-dimensional gene expression data. In *Cladag 2019 - Book of Short Papers*, pages 132–135. EUC, Cassino.

Coraggio, L. and P. Coretto (2019). Likelihood-type methods for comparing clustering solutions. In *Cladag 2019 - Book of Short Papers*, pages 120–123. EUC, Cassino,.